

# Rossmel Abasto

Frontend | Fullstack Developer

Cochabamba, Bolivia | [abastorossmel@gmail.com](mailto:abastorossmel@gmail.com) | [linkedin.com/in/rossmel/](https://www.linkedin.com/in/rossmel/) | [github.com/rossmelabasto](https://github.com/rossmelabasto) | [portfolio.rossmel.top](https://portfolio.rossmel.top)

## PROFESSIONAL SUMMARY

Frontend Developer with 3+ years of experience at WANT Digital Agency building multi-tenant SaaS, mobile apps and websites with React, Next.js, React Native and TypeScript (~1,800 commits across client products in Bolivia and the US). Solid programming foundation: self-taught since 2020, with years of writing production code without AI assistants. Growing into Fullstack with FastAPI, Node.js, SQLite/PostgreSQL, Firebase and Supabase. Since 2025 I integrate AI into products (hybrid RAG, local and cloud LLMs) and into my daily workflow, to ship faster without sacrificing quality.

## TECHNICAL SKILLS

**Frontend & Mobile:** React, Next.js (Pages & App Router), React Native, Expo, TypeScript, JavaScript, HTML5, CSS3 / Sass, Tailwind CSS, MUI, shadcn/ui, Redux Toolkit, i18n, PWA, Astro, GSAP, Motion

**Backend & Data:** Node.js, Express, Python, FastAPI, REST APIs, SSE, NextAuth, SQLite (FTS5, WAL), PostgreSQL, Prisma, Drizzle, Supabase, Firebase, Electron

**Applied AI:** RAG, Hybrid search (BM25 + embeddings), LLM evaluation, Ollama, Groq, Gemini, OpenAI API, DeepSeek, sqlite-vec, GitHub Copilot, Claude Code, AI agents

**DevOps, Quality & Tools:** Git, GitHub Actions, GitLab, Docker, Linux (Arch), systemd, Cloudflare Tunnel, Tailscale, KVM / libvirt, pytest, Vitest, Vercel, Bruno, Figma

## WORK EXPERIENCE

### Fullstack Developer — Independent

Jan 2026 – Present | Cochabamba, Bolivia | Freelance

- Link'u (paid project): offline-first desktop app for a growing rural community's water billing (Electron, React, SQLite), with AES-256-GCM encrypted backups, auto-updates, 47 tests and cross-platform CI.
- SGPG: thesis project management system for UDABOL, used by the head of the Systems Engineering program (Next.js 16, Firebase), with AI-powered PDF summary and tag extraction (Groq), document versioning and audit trail.

### Frontend Developer (Web & Mobile) — WANT Digital Agency / GeekLabs

Aug 2022 – Jan 2026 | Cochabamba, Bolivia | Internship — Contractor — Employee (2025–2026)

- Lead frontend developer of Vulcano (now CarX), GeekLabs' multi-tenant SaaS for auto repair shops, piloted at a real shop (944 of 1,131 commits): work orders, quotes, vehicles, revenue reports, calendar, CSV import, role-based dashboards and ES/EN i18n with Next.js, TypeScript, MUI, Redux Toolkit and NextAuth.
- Wired the frontend to Vulcano's AI module (built by the tech lead) that auto-generates the services of a work order.
- Built React Native/Expo mobile apps: Adsie, an ads network for Florida connecting local businesses, customers and influencers (434 of 497 commits); Vulcano's mechanic app (push notifications, service timer); One Life Fitness (a gym in Cochabamba and La Paz) and OneHand (physiotherapy appointments).
- Built the frontend of Bite, a multi-tenant digital menu and ordering SaaS for restaurants (Next.js App Router): QR table ordering, checkout, coupons, branches, plans and Google Maps.
- Built the WordPress website for LYNX Bolivia, an electronics and home appliance retailer.
- Worked alongside the tech lead (backend) integrating REST APIs, with code reviews and short delivery cycles under tight deadlines.
- Started as an intern (3 months), continued as a monthly-paid contractor and was formally employed from 2025 to 2026.

## PROJECTS

### AdvAI — Capstone · UDABOL (2026)

AI legal contract auditing with verifiable citations. Hybrid search: BM25 (SQLite FTS5) + bge-m3 embeddings fused with RRF over 3,267 articles and 971 supreme court rulings — retrieval hit@8 went from 42.9% to 80%. The model only picks keys from the sources shown and the quoted text is inserted by the system: 82.9% correct citations and 94.6% of clauses with no severe error in internal evaluation.

Technologies: Next.js 16, React 19, TypeScript, Python, FastAPI, SQLite FTS5, Ollama, RAG, SSE, pytest

## **Homelab — Personal project · Infrastructure (2026)**

A 24/7 home server running a dozen services, operated by an AI agent (OpenClaw). Secure exposure via Cloudflare Tunnel (no open ports), private access over Tailscale, UFW and fail2ban. Services on Docker and systemd: media server, home management, a custom RAG notebook and more; timer-based backups.

Technologies: Arch Linux, Docker, systemd, Cloudflare Tunnel, Tailscale, OpenClaw, KVM

## **Notebook — Personal project · On my server (2026)**

Study notes with a chat that answers by citing your own notes. Custom RAG: notes are split into chunks, embedded and searched by similarity with sqlite-vec inside SQLite. No separate vector database. Answers with sources: only those above a threshold calibrated on real data are shown, and each citation links to the exact text in the note.

Technologies: Node.js, Express, SQLite, sqlite-vec, RAG, Embeddings, DeepSeek, JavaScript, systemd

## **Rubik — Personal project · On my server (2026)**

Practice the Rubik's cube and other puzzles, with official scrambles and a 3D viewer. Official scrambles and a 3D viewer with cubing.js, the cubing community's library, including its WebAssembly scramble generator. No CDN dependency: the library ships with the site itself, so it keeps working if an external service goes down.

Technologies: JavaScript, cubing.js, WebAssembly, HTML, CSS

## **rOS — Personal project · Linux (2026)**

My own flavor of Arch, daily-driven and on its way to becoming a distro. Wayland desktop with niri and DankMaterialShell (Quickshell/QML); greetd greeter; Limine boot with custom branding. ros-update: Timeshift snapshot before updating pacman, AUR and flatpak; GPU mode switching (PRIME) and automatic power profiles.

Technologies: Arch Linux, niri, DankMaterialShell, Limine, Shell, systemd

## **EDUCATION**

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### **B.S. Systems Engineering — Universidad de Aquino Bolivia (UDABOL)**

2021 – 2027 (expected) · Final semester; internal defense Dec 2026, external Mar 2027

Capstone: AdvAI — AI-powered legal contract auditing (hybrid RAG + LLMs).

### **Postgraduate Diploma in Software Development — Universidad de Aquino Bolivia (UDABOL)**

2026 – in progress · 4 modules

### **Web Application Development — Solaning**

2022 · Completed

**Additional Training:** Platzi, Coursera, Udemy, freeCodeCamp, Google for Education, W3Schools

## **LANGUAGES**

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Spanish (Native), English (Advanced (C1))

## **SOFT SKILLS**

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Problem solving, Autonomy, Working under pressure, Fast learner, Client communication, Technical writing